

Aaron Espinoza

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SYSTEMS EXPERIENCE

Open Source Contributor

Sept. 2025 – Present

The FreeBSD Project

Remote

- Fixed a kernel panic in `if_bridge(4)`: a NULL `softc` dereference on the mbuf pullup failure path in `bridge_input()`. Found while tracing the driver during early `veb(4)` work; reviewed and landed through the project's code review process. Commit history
- Developing `veb(4)` (see Projects), a virtual Ethernet bridge pseudo-device built on `if_clone(9)` and the `ifnet` KPI: stripped down relative to `if_bridge(4)` and hidden from the host network stack, cutting per-packet overhead for latency-sensitive jail and bhyve networking. Design review targeted at `freebsd-net@`.
- Maintain RustKLD, including CI, test coverage, and contributor documentation. Active in driver and networking discussions on the project mailing lists.

Google Summer of Code Contributor

May 2025 – Sept. 2025

The FreeBSD Project

Remote

- Delivered a modular Rust framework for FreeBSD kernel device drivers with safe wrappers over C kernel objects (devfs nodes, character devices, memory allocation), removing the need for contributors to hand-roll FFI per driver.
- Wrote 11 automated ATF/Kyua tests covering stress, concurrency, error paths, and memory leaks; integrated into CI so regressions surface on every change.
- Fixed devfs registration, quiesce/unload handling, and LLVM 19 compatibility in existing Rust drivers, restoring a stale build against CURRENT; shipped reference drivers and documentation now used by new contributors.
- Project writeup | Official FreeBSD project video

Maintenance Technician

Aug. 2022 – May 2024

tekRESCUE

San Marcos, TX

- Ran weekly maintenance across client Windows and Linux fleets; specified and built custom servers and handled on-site hardware and network troubleshooting.

SYSTEMS PROJECTS

veb(4) — Virtual Ethernet Bridge for FreeBSD

In progress

Port of OpenBSD's Layer 2 bridge pseudo-device to the FreeBSD network stack. Out-of-tree KLD build with CI running `kldload` and `vnet` jail tests, plus drift detection against a `freebsd-src` branch. *C, FreeBSD, if_clone(9), ifnet KPI, ATF/Kyua, Cirrus CI.*

RustKLD — Rust Kernel Driver Framework for FreeBSD

github.com/Acesp25/rustkld

Modular framework and test harness for FreeBSD kernel loadable drivers written in Rust: safe abstractions over C kernel interfaces, CI integration, and contributor documentation. *Rust, C, FreeBSD, ATF/Kyua.*

VirtIO GPIO Driver

In progress

FreeBSD general-purpose I/O driver written in C against the OASIS VirtIO 1.3 specification; implements the core device and queue negotiation path, IRQ support in progress. *C, FreeBSD, VirtIO, QEMU.*

EDUCATION

Texas State University

Expected Dec. 2026

B.S. Honors Computer Science; Minors in Applied Mathematics and Chinese

San Marcos, TX

- GPA 3.98/4.00 overall, 4.00/4.00 College of Science & Engineering GPA. Dean's List/President's List, all semesters.
- Extracurriculars: Vice President, TXST Fencing Club (40+ members), TXST Esports Club CS2 Captain.

RESEARCH & PUBLICATION

Undergraduate Deep Learning Researcher, Texas State University

Aug. 2024 – Present

- GAN-based adversarial perturbation research against facial recognition; 98.7% average impersonation success on LFW and CASIA-WebFace. First-author paper under review at AAAI 2027.
- Placed 1st globally in the MICCAI 2024 WHS++ segmentation challenge as an AI research intern at Guangdong Provincial People's Hospital (May–Aug. 2024).
- *Enhance Multi-modal and Multi-center Whole Heart Segmentation Using Data Augmentation and Model Calibration.* CARE Workshop, MICCAI 2024. doi.org/10.1007/978-3-031-87009-5_13

SKILLS

Systems: FreeBSD kernel and device drivers, KLDs, devfs, VirtIO, network pseudo-devices, POSIX (pthreads, sockets, IPC)

Languages: C, Rust, Python, Bash, Assembly (MIPS, x86-64), Java, C++

Networking: Layer 2 bridging, TCP/IP, pf, WireGuard, NFS, unbound

Virtualization & Tooling: bhyve, QEMU, jails, ATF/Kyua, DTrace, GDB, Wireshark, CI pipelines, Git, Vim, VS Code